Week 4 | Acquisition Requirements

AUSCULTAFORGE - Smart Digital Stethoscope | EEE 495
Module A: Acquisition Hardware & Characterisation | Kaan
19 October 2026 | Revision 1.1 | Requirements baseline and Week 5 preparation

This work is a student project and makes no diagnostic claim. Physical validation is restricted to an acoustic phantom. No human recordings are planned. Physical performance: **NOT YET MEASURED.** [1,2]

1) Scope, progress and evidence status

The advisor's Week 4 Module A task was mapped to four deliverables: gain, bandwidth, noise floor and clipping margin. The coordination brief dated 17 October was reviewed alongside this task. It adds the INMP441 main path, an ECM comparison path and the first phantom specification. Requirements and verification procedures were documented below; component arrival, tool installation and test completion have not been evidenced. [1-4]

Change from Week 3: The PUI DMM-4026-B-I2S-R microphone with its DMM-4026-B-I2S-EB-R evaluation board, selected in Week 3, could not be obtained within the required timeframe because of the supplier lead time. INMP441 microphone modules were therefore purchased instead. INMP441 was designated as the main integration path in the coordination brief. Kingstate was retained for comparison. The earlier SPH0645 discussion did not become the reported purchase. This is an architecture update, not a measured final-sensor decision. [4,9]

Reported order / allocation	Qty.	Technical role and remaining evidence
INMP441 modules	3	Main digital path. Exact board variant, pin map and incoming inspection remain open.
Kingstate KECG2740PBJ	2	Analogue comparison candidate; external conditioning and digitisation still required.
MCP6022 op-amps	5	ECM conditioning parts; package, circuit values and ADC remain to be confirmed.
Exciter, 30 mm / 12 ohm	2	Phantom driver; manufacturer, power rating and response are not supplied.
PAM8403 modules	4	Phantom drive amplifiers; board layout and input capacitors require inspection.
Breadboard / jumpers / prototyping items	As repo rted	Assembly support. Exact inventory and receipt status remain to be checked.

Order placement was confirmed by Kaan; the date of 16 October and quantities above were taken from the coordination brief. Receipt, delivery and physical inspection are not inferred from an order. Supplier details and receipts must be linked to the shared BOM. [4]

Baseline status

The main-path direction and verification methods are documented. Numerical operating targets marked **PROPOSED** are Module A engineering proposals for joint review, not advisor-issued limits. Full requirements freeze remains conditional on the open decisions in Section 7. No hardware PASS or Week 4/5 gate completion is claimed.

Measurable Requirements

Week 4 | The four items explicitly required by the advisor

2) Requirements and acceptance evidence

The 20-500 Hz investigation band is retained. The 20-150 Hz emphasis in the component document and the known INMP441 low-frequency limitation are recorded together. The band is not silently reduced to match the purchased microphone. [1,3,5]

ID / subject	Requirement or proposed operating target	Verification / closure condition
A4-01 Bandwidth	Characterise the entire 20-500 Hz path; explicitly report 20, 30, 40, 50, 60, 80, 100, 150, 200, 300 and 500 Hz. Preserve unfiltered acquisition.	Measure response against the logged drive and an independent reference where available. Report loss relative to 100 Hz and the reference conditions. Allowable ripple/loss: OPEN for advisor review.
A4-02 Gain	INMP441: raw capture scaling = 1 (0 dB software gain); no automatic gain control or normalisation in the raw record. ECM: select and record gain from input range and ADC headroom.	Confirm signed sample decoding. Measure ECM Vout/Vin with a known electrical input before comparing acoustic records. Absolute ECM gain value: OPEN pending ADC/input envelope.
A4-03 Noise floor	Record three 30 s runs per baseline condition; calculate unweighted RMS noise over 20-500 Hz and separately 20-60 Hz and 60-500 Hz.	Store raw data, PSD, processing settings and run-to-run spread. Absolute acceptable noise / minimum useful signal: OPEN; catalogue dBA is not an acceptance limit.
A4-04 Clipping margin	PROPOSED: retain at least 6 dB peak headroom at the declared maximum normal phantom stimulus; zero saturated samples in normal tests.	Measure raw peak level and clip count; repeat at stepped drive levels. Controlled overload tests must be separately labelled. Team/advisor confirmation of 6 dB target is pending.
A4-05 Dynamic range	Report the span from measured band-limited noise to the maximum accepted clean output at fixed settings.	Record waveform, harmonics and level compression. Distortion acceptance limit remains OPEN; a 24-bit word is not a measured dynamic range.
A4-06 Repeatability	Use the same source, drive, fixture, contact load and cable layout for both paths. Initially perform three independent remounts.	Preserve all repeats and report spread. Extend to the advisor's at-least-ten-repeat campaign in Week 11.

Freeze boundary: a test method is specified even where a performance limit remains open. These open limits prevent an unconditional freeze claim. They must be resolved with the team/advisor before the corresponding acceptance test; observed results must not be used to move a target silently. [2]

Required record for every run

Run ID; date; operator; module/unit ID; circuit and fixture revision; firmware/script version; source file and hash; sample rate/format; gain; supply voltages; stimulus setting; coupling thickness; clamp force or measured surrogate; ambient conditions; raw filename; processing configuration; result status. Ege owns the final metadata schema; these are Module A input requirements.

INMP441 Main Acquisition Path

Week 4 | Manufacturer evidence and low-frequency risk

3) Device specification and interface handoff

INMP441 manufacturer data [5]	Published value / condition
Supply / sensitivity	1.62-3.63 V operating range; -26 dBFS typical at 1 kHz, 94 dB SPL, tolerance +/-3 dB.
Frequency limits	Typical acoustic lower -3 dB point: 60 Hz; upper point: 15 kHz.
Noise / dynamic range	Typical 61 dBA SNR; -87 dBFS A-weighted noise over 20 Hz-20 kHz; 87 dB dynamic range. Not measured project values.
Digital output	24-bit signed two's-complement I2S; 64 bit clocks per stereo frame; left/right selection by L/R.

The internal ADC removes the external microphone AFE/ADC requirement on this path. At 48 kHz, the separate internal digital high-pass corner is 3.7 Hz; this does not remove the acoustic 60 Hz limitation. [5]

Proposed A/B operating profile: regulated 3.3 V; 48 kHz sampling; two 32-bit I2S slots with one active microphone channel; BCLK = 64 x 48,000 = 3.072 MHz. This follows the earlier REV01 direction but awaits Ozan's current interface confirmation. GPIO numbers, USB framing and packet layout are not assigned by this report. [10]

At incoming inspection, the module pinout, port opening, local decoupling and enable wiring shall be checked. The chip recommendation includes 100 nF local bypassing. I2S bit alignment, sign extension and inactive-slot handling shall be verified with Module B. A 32-bit transport container must not be reported as 32-bit sensor resolution. [5]

Low-frequency consequence and calculation

A **first-order high-pass approximation** was calculated only to illustrate the risk: $|H(f)| = f / \sqrt{(f^2 + 60^2)}$. It is not a fitted INMP441 model or a physical measurement. Real mounted response must be measured.

Frequency (Hz)	20	30	40	60	100	500
Illustrative loss (dB)	-10.00	-6.99	-5.12	-3.01	-1.34	-0.06

Attenuation in this region can reduce low-frequency S1/S2 energy and change waveform shape, amplitude ratios and quality metrics. Recognisable sounds alone do not establish full-band fidelity. Digital boost cannot recover information already buried in noise; any equalised record must be a separately labelled derivative of preserved raw data.

Reason for the PUI change

The Week 3 PUI candidate could not be obtained within the required timeframe because of the supplier lead time. INMP441 modules were therefore purchased as the replacement, and the change was adopted as the current integration basis. This procurement reason was confirmed by Kaan on 19 October 2026. The explicit 60 Hz risk is now a tracked deviation. Continued use depends on phantom evidence and advisor review, rather than treating procurement as proof of suitability. [4,9]

ECM Path and Measurement Definitions

Week 4 | Gain, noise and clipping are separate engineering quantities

4) Kingstate comparison path

Planned chain: **Kingstate - bias network - MCP6022 conditioning - anti-aliasing / ADC input interface - ADC or suitable USB audio interface - recorded samples.** The ADC route, supply/reference levels, gain values and anti-aliasing design remain open with Ozan. No new analogue circuit was built for this report. [4]

Kingstate specifies 2 V standard operation with a 2.2 kohm test load and -40 +/-3 dB re 1 V/Pa sensitivity at 1 kHz: nominally 10 mV/Pa. Its listed 20 Hz lower range does not specify an explicit lower -3 dB point. MCP6022 supports 2.5-5.5 V operation and 10 MHz gain-bandwidth; it is an op-amp, not an ADC. Purchased package and actual board behaviour must be checked. [6,7]

Gain selection rule: measure the maximum expected ECM signal at the selected bias. Choose gain G so G x $V_{in,peak}$ remains below 0.501 x $V_{allow,peak}$ for the proposed 6 dB margin, including DC offsets and the tighter of op-amp swing and ADC limits. Document every high-pass pole; coupling capacitors must not unintentionally remove the 20-60 Hz content. Final resistor/capacitor values require the chosen ADC and measured input envelope.

Noise procedures

Digital main path: record with the exciter off and the microphone mounted in the final fixture. Repeat with the amplifier powered but silent, then with it powered off. Keep environment and settings fixed. This is a system quiet-condition baseline, including room and mechanical noise; it is not an isolated microphone self-noise measurement. Do not short I2S outputs or power pins.

ECM electronics: disconnect the capsule and terminate the analogue signal input at the correct AC reference while maintaining its DC bias. Measure shorted-input electronics noise, then repeat with representative source impedance, and finally with the capsule installed in a quiet fixture. Never short a supply rail. If only an oscilloscope is used initially, record its own noise and bandwidth; do not claim a noise level below the instrument capability.

Calculation and clipping conventions

After correct signed decoding, define $x[n] = \text{sample} / 2^{23}$ for INMP441. Use **peak dBFS = 20 log10(max|x|)**. For this report, RMS dBFS is explicitly referenced to a full-scale sine: **20 log10(RMS(x) / (1/sqrt(2)))**. Report mean/DC separately; retain untouched raw samples. Noise RMS is calculated from the integrated PSD over each stated band. Window, segment length and calibration must be stored.

Peak headroom is -peak dBFS; the proposed 6 dB target corresponds to max|x| <= 0.501. Count exact digital rail hits and inspect level compression/harmonics because analogue or acoustic overload can occur before code saturation. For ECM, report voltage headroom at each stage as well as ADC code headroom.

For a fixed phantom signal, compare signal/noise power using matched source-on and source-off runs, with noise-power subtraction only when the estimate is positive and stationary. Report both raw-level and gain-normalised comparisons; do not confuse normalised waveform similarity with sensitivity. Equivalent input noise in Pa or SPL requires independent acoustic calibration.

Phantom Drive Requirements

Week 4 | Safe electrical operation and an identifiable test method

5) Exciter and amplifier assessment

The reported 30 mm / 12 ohm exciters and PAM8403 modules were assigned to the phantom. **PAM8403 is the exciter power amplifier, not the microphone AFE.** The exact exciter rating and module schematic remain missing. Therefore electrical compatibility is provisional and mechanical output at 20 Hz is not guaranteed. [4]

PAM8403 is a bridge-output class-D amplifier with a 2.5-5.5 V operating supply. Its advertised approximately 3 W output is specified under a 4 ohm / 5 V / high-distortion condition and cannot be transferred to the 12 ohm exciter. The module's input coupling capacitor also affects low-frequency drive. [8]

Requirement	Planned implementation / check
Power and load	Use a regulated, current-limited bench supply, initially at 3.3 V and low drive; increase only within verified module/exciter ratings. Do not power the exciter from the microphone supply or a GPIO.
Bridge-output wiring	Connect one exciter across the + and - outputs of one channel. Neither output is ground. Do not parallel channels. Connect the load with power off.
Measurement connection	Use an appropriate differential measurement across the output. Do not attach an earth-grounded oscilloscope clip to a switching output. Account for PWM when measuring audio RMS.
Low-frequency drive	Inspect input capacitors and effective input resistance; assess f_c = 1/(2 pi R C). Verify loaded audio-band response rather than relying on the module name.
Exciter limits	Obtain its manufacturer/model, continuous power, mounting instructions and displacement limits. Log audio differential voltage, supply current and temperature behaviour during gradual tests.
Interference control	Keep amplifier/exciter wiring away from the microphone and I2S cable; use short paired drive wires and strain relief. Compare powered-silent and power-off noise runs.

Planned 20-500 Hz test sequence

1. Verify polarity, continuity, mounting and supply limits before energising. Begin with low-level excitation; stop for severe distortion, abnormal heating, rubbing or current-limit operation.
2. Apply stepped sine tones at the A4-01 frequencies. Proposed record length: 5 s per tone after at least 1 s settling, repeated three times. Log generator/DAC level and loaded drive voltage.
3. Record microphone output and, when available, an independently characterised acoustic or vibration reference at the coupling surface.
4. Report the complete fixture transfer first. Constant electrical voltage does not imply constant acoustic pressure; without a reference, sensor-only response and absolute SPL cannot be claimed.
5. Replay the team's already selected PCG record at a fixed level; preserve source identity and compare S1/S2 structure, band energy and noise with Module C.

A DAC or verified analogue audio output is needed between the PC playback stream and PAM8403. The amplifier does not accept I2S directly. Ege provides the playback source; the available output device and cables must be confirmed.

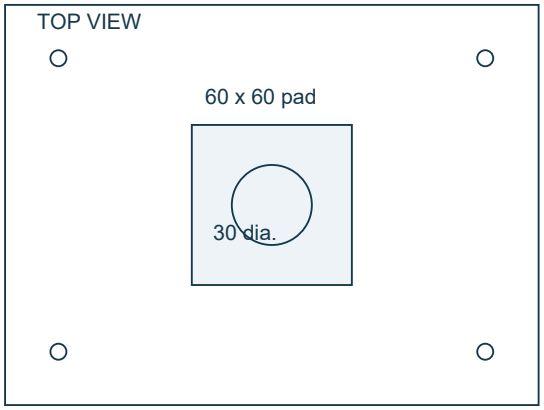
Mechanical Phantom Specification

Week 4 | Initial buildable proposal; dimensions require fit verification

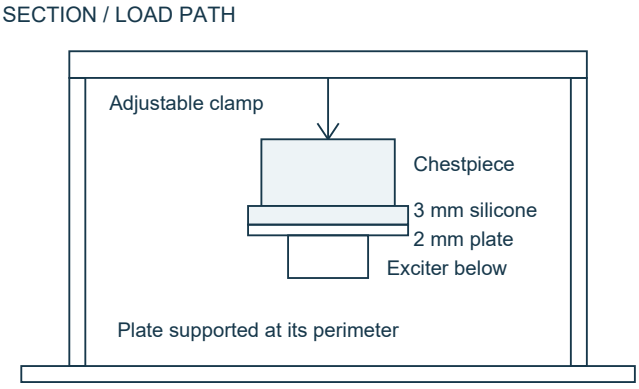
6) Coupling, frame and interchangeable mounting

A **dry silicone sheet, nominal 3 mm thick**, was proposed as the first coupling material. Begin with a 60 x 60 mm coupon. Record supplier, hardness if available, measured thickness and compression. Gel was retained as a later alternative; it must not enter the microphone port. These are starting design choices, not measured optimal dimensions.

PROPOSED FIXTURE - dimensions in mm; schematic, not to scale



Base: 200 x 150 x 6



Standoffs: 80 mm nominal; clamp travel: 20 mm

Part / feature	Initial specification and verification
Rigid support	200 x 150 x 6 mm plywood base; 160 x 30 x 10 mm bridge; nominal 80 mm standoffs. Four mounting centres at (20,20), (180,20), (20,130), (180,130) mm from one base corner. Confirm fastener/hole fit before cutting.
Driven surface	60 x 60 x 2 mm acrylic plate, perimeter-supported; exciter bonded centrally underneath according to its mounting instructions. Silicone coupon above. Plate response is part of the measured phantom.
Clamp and load	At least 20 mm adjustment travel. PROPOSED initial normal load: 1 N, checked with a force gauge or calibrated scale arrangement. Measure actual contact area before reporting pressure; 1 N is not a validated optimum.
Interchangeable holder	Use the same chestpiece and defined cavity with removable sensor adapters. Measure the actual chestpiece/module first; record cavity depth, port position, seal geometry and any unavoidable adapter differences.
Port and cables	Align the bottom acoustic port with an unobstructed opening; seal around it without covering it. Support the PCB rather than the microphone can. Fix the cable to the frame so its weight does not change contact force.

Mechanical acceptance: no rocking, port blockage or cable-induced lifting; reproducible repositioning against physical stops. Log clamp setting and measured load before every run. Quantitative remounting tolerance will be agreed after initial fixture checks and before the formal comparison. Silicone, plate/frame material, gasket, clamp, fasteners and force-measurement access are not confirmed as procured.

Week 5 Handoff and Open Decisions

19-25 October 2026 | Readiness plan; no completed experiment is implied

7) Work sequence and responsibility

Owner / timing	Planned work	Required evidence
Kaan / before arrival	Prepare fixture drawing, run log, source-controlled analysis procedure and lab bookings. Confirm tools and inventory.	Drawing revision; equipment-access list; analysis environment/version record.
Kaan + Ozan / on arrival	Photograph and identify all modules; verify supplies/pins, then connect INMP441 safely and confirm I2S decoding.	Unit IDs; connection record; measured supply/clock values; first raw file.
Kaan / bench work	Construct fixture and check PAM8403/exciter at low drive. Begin noise and controlled-tone tests when the recording path is available.	Raw data and metadata; script-generated noise/headroom summaries; no PASS before evidence.
Kaan + Ozan / ECM	Agree ADC route and initial breadboard conditioning; plan safe input termination and electronics-noise measurement.	Circuit, gain/bias choices, termination method and ADC decision.
Ozan + Ege / shared gate	Verify synthetic MCU-to-PC transfer, live view and saved records. Kaan provides acquisition conditions and hardware status.	Known-input end-to-end record, sample continuity checks and component receipt evidence.

Advisor-plan conflict: Week 5 explicitly requires a breadboard front end and shorted-input noise measurement. A digital microphone has no accessible analogue input for that test. The ECM branch can address the original task; if it is deferred, a documented advisor-approved alternative is required. The coordination brief does not cancel this requirement. The assessment gate also requires components/data in hand and the first end-to-end signal. A PDF or a synthetic stream alone does not demonstrate every Module A requirement. [1,2,4]

Open item	Owner / closure evidence
LF suitability and freeze limits	Kaan + advisor: acceptable 20-60 Hz loss, response tolerance, minimum useful signal/noise and distortion limits; approve or reject the proposed headroom target.
Module B interface	Ozan + Kaan: current MCU variant, GPIO map, I2S profile and ECM ADC/reference. No teammate's unconfirmed decision is overwritten.
Missing mechanical / playback items	Kaan + team: silicone, frame/plate, gasket, chestpiece/adapter, clamp, fasteners, strain relief; DAC/audio output and suitable cables. Verify inventory before buying.
Lab access and electrical ratings	Kaan: oscilloscope/differential measurement, generator, multimeter, supply, soldering, force check; exciter power rating and actual amplifier module parts.
Procurement and reproducibility	Team: receipts, exact part/board numbers, arrival status; tool installation, fixed dataset identity and reproducible script outputs.

Physical gain, frequency response, noise, clipping, dynamic range and repeatability: **NOT YET MEASURED.** Week 2 and Week 3 records remain unchanged. The new architecture and its limitations belong in the Week 4 revision history.

Traceability and References

Week 4 | Evidence basis and limits of this submission

8) Requirement-to-report mapping

Source requirement	Report coverage
Advisor Week 4: gain	A4-02; main-path raw scaling; ECM gain/headroom rule in Section 4.
Advisor Week 4: bandwidth	A4-01; 60 Hz acoustic corner and explicitly labelled calculation in Section 3.
Advisor Week 4: noise floor	A4-03; digital quiet baseline and ECM terminated-input procedures in Section 4.
Advisor Week 4: clipping margin	A4-04; proposed 6 dB margin, raw peak/rail/compression checks in Section 4.
Coordination additions	Reported order and architecture change; ECM dependencies; phantom electrical/mechanical requirements; Week 5 handoff and open decisions.
Assessment: verifiable evidence	Run metadata, independent-reference limitation, script-generated results, separate proposal/result status and preserved history.

References

[1] Advisor. SDP4_Digital_Stethoscope_Plan_TR-1.docx, Section 7: Semester I Weeks 4-5; scope and comparison requirements. Preserved project source.

[2] Advisor. SDP_Assessment_Criteria_for_Students.docx: requirement specification, individual evidence, gates and common rules. Preserved project source.

[3] Advisor. SDP4_Component_Requirements_EN.docx: Group A/C and Notes Before Ordering, especially low-frequency evidence and laboratory inventory.

[4] AuscultaForge Technical Coordination. Module A - Week 4 Closure and Week 5 Preparation, 17 October 2026; supplied by Kaan on 18 October. Order placement additionally confirmed in Kaan's message; the supplier-lead-time reason for replacing PUI with INMP441 was confirmed on 19 October. No receipt or delivery inspection was supplied.

[5] InvenSense/TDK. INMP441, DS-INMP441-00, Rev. 1.1, pp. 4-5, 8, 11-15. Manufacturer datasheet.

[6] Kingstate. KECG2740PBJ specification, pp. 2-3. Manufacturer datasheet hosted by Farnell.

[7] Microchip. MCP6021/1R/2/3/4, DS20001685E. Manufacturer datasheet.

[8] Diodes Incorporated. PAM8403, November 2012, pp. 1-4 and 8-9. Manufacturer datasheet hosted by Mouser.

[9] Kaan. Week 3 Module A Candidate Selection v2, numbered English report, 6 October 2026; archived in Week 3 PR #9. PUI was the earlier comparison candidate.

[10] Ozan. REV01 System Design, earlier interface proposal. Its 48 kHz / ESP32-S3 direction is used only as a proposed interface basis; current Module B sign-off remains required.

External technical references were checked on 18 October 2026. Manufacturer values describe the specified devices under their stated conditions; they are not acceptance evidence for the purchased breakout boards. Square-bracket numbers identify the references above.

Submission boundary

This report documents Module A requirements and test preparation. Hardware verification, full team freeze and advisor acceptance remain pending. The procurement change does not establish low-frequency compliance. All derived illustration values were generated by the accompanying calculation script; no experimental numbers were inserted.